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AgroSmart Marketplace: A Smart Agriculture Trading Platform

Software Requirements Specification (SRS)

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1. Introduction

1.1. Purpose of this Document

The purpose of this Software Requirements Specification (SRS) document is to clearly define the requirements of the AgroSmart Marketplace system. This document describes the system's objectives, functional requirements, interface requirements, design constraints, and non-functional attributes that are required to develop the proposed web-based agriculture trading platform.

AgroSmart Marketplace is designed to support direct trading between farmers and buyers by reducing the dependency on intermediaries. The system allows farmers to publish agricultural products and enables buyers to search for products, view product details, submit price offers, and interact with farmers through a structured negotiation-based process. The system also includes location-based matching, demand analysis, smart pricing support, and review and feedback features to improve transparency, trust, and decision-making.

This document will be used as a guide for system development, testing, evaluation, and future maintenance. It helps developers understand what the system should perform, assists testers in validating system functionality, and provides the project supervisor and evaluators with a clear understanding of the expected system behavior and scope.

1.2. Scope of the System

The scope of the AgroSmart Marketplace system is to provide a web-based smart agriculture trading platform that supports direct interaction between farmers and buyers. The system is designed to reduce the dependency on intermediaries and improve transparency, fairness, and efficiency in agricultural trading activities.

The system allows farmers to register, log in, and publish their agricultural products with relevant details such as product name, category, price, minimum price, quantity, location, and product images. Buyers can register and log in to the system, browse available agricultural products, search for products, view product details, and submit price offers for selected products.

The system supports a negotiation-based trading process where farmers can review buyer offers and make decisions such as accepting, rejecting, or responding with counter-offers.

The system also includes location-based matching, which helps buyers find nearby farmers and products based on district and city information. In addition, the system provides demand analysis features to identify high-demand and low-demand products based on buyer activities such as searches, views, offers, and orders. These insights support better decision-making for both farmers and buyers.

The main user roles of the system are Buyer, Farmer, and Admin. Buyers interact with products and submit offers, farmers manage products and offers, and administrators manage system users, product information, and overall platform activities. The system also includes review and feedback features to improve trust between users.

The current scope of the system does not include online payment integration, delivery management, advanced machine learning prediction, or a dedicated mobile application. These features may be considered as future enhancements.

1.3. Intended Audience

The intended audience of this Software Requirements Specification (SRS) document includes the project supervisor, academic evaluators, system developer, testers, and future maintainers of the AgroSmart Marketplace system. This document provides a clear understanding of the system requirements, expected functionalities, constraints, and quality attributes.

The project supervisor and academic evaluators can use this document to review and evaluate whether the system meets the project objectives. The developer can use it as a guideline for system design, implementation, and testing, while testers can use it to verify the functional and non-functional requirements.

The main end users of the AgroSmart Marketplace system are farmers, buyers, and administrators. Farmers use the system to publish and manage agricultural products, buyers use the system to browse products and submit offers, and administrators use the system to manage users and monitor platform activities.

1.4. Definitions, Acronyms, and Abbreviations

- SRS - Software Requirements Specification. A formal document that defines the requirements, functions, constraints, and expected behavior of the software system.
- AgroSmart Marketplace - A web-based smart agriculture trading platform that connects farmers and buyers directly.
- Buyer - A registered user who can browse agricultural products, view product details, and submit price offers.
- Farmer - A registered user who can publish agricultural products, manage product details, and review buyer offers.
- Admin - A system user responsible for managing users, products, and overall platform activities.
- UI - User Interface. The visual part of the system that allows users to interact with the application.
- UX - User Experience. The overall experience and satisfaction of users when using the system.
- CRUD - Create, Read, Update, Delete. Basic database operations used to manage system data.
- Database - A structured collection of data used to store users, products, offers, orders, and reviews.
- Demand Analysis – A system feature that analyzes buyer activities such as searches, views, offers, and orders to identify high-demand and low-demand products.
- Smart Pricing - A feature that supports price suggestions based on demand level, market conditions, and system rules.
- Location-Based Matching - A feature that connects buyers with nearby farmers based on district and city information.
- Negotiation System - A feature that allows buyers to submit offers and allows farmers to accept, reject, or respond with counter-offers.

2. General Description

2.1. Product Perspective

AgroSmart Marketplace is a web-based agriculture trading platform developed to support direct interaction between farmers and buyers. The system functions as an independent marketplace application where farmers can publish agricultural products and buyers can browse products, view details, and submit price offers through a structured negotiation process.

The system reduces dependency on intermediaries by providing a digital platform for agricultural trading. Farmers can manage product details, prices, minimum price values, and buyer offers, while buyers can search for products, identify nearby farmers, and submit offers based on their requirements.

AgroSmart Marketplace also supports location-based matching, demand analysis, smart pricing support, and review and feedback features to improve transparency, trust, and decision-making. From a system perspective, it consists of a frontend user interface, backend application logic, and a database that stores users, products, offers, orders, reviews, and activity-related data.

2.2. Product Functions

The main functions of the AgroSmart Marketplace system are as follows:

- **User Registration and Login -** The system allows buyers and farmers to register and log in securely according to their user roles. Administrators can log in using accounts created by the system owner or authorized personnel.
- **Role-Based Dashboard Access -** The system redirects users to separate dashboards based on their roles, such as Buyer, Farmer, and Admin.
- **Product Publishing -** Farmers can publish agricultural products by adding details such as product name, category, price, minimum price, quantity, location, and product images.
- **Product Browsing -** Buyers can browse available agricultural products and view product information such as price, farmer details, location, and demand level.

- **Offer Submission -** Buyers can submit price offers for selected products by entering an offer price, quantity, and optional note.
- **Negotiation Process -** Farmers can review buyer offers and make decisions such as accepting, rejecting, or responding with counter-offers.
- **Location-Based Matching -** The system helps buyers find nearby farmers and products using district and city information.
- **Demand Analysis -** The system identifies high-demand and low-demand products based on buyer activities such as searches, views, offers, and orders.
- **Smart Pricing Support -** The system supports better pricing decisions by considering demand levels, minimum prices, and market-related information.
- **Review and Feedback Management** - Buyers can provide reviews and feedback after completed orders to improve trust and product reliability.
- **Order Management -** The system manages order-related activities after offers are accepted by farmers.
- **Admin Management -** The administrator can manage users, products, and overall platform activities to maintain system reliability.

2.3. User Classes and Characteristics

User Class	Description	Characteristics
Buyer	Registered users who browse agricultural products, view product details, and submit price offers.	Basic computer knowledge, web browsing skills, internet access
Farmer	Registered users who publish agricultural products, manage product details, pricing, and buyer offers.	Basic web application knowledge, internet access, understanding of product pricing
Administrator	Authorized users responsible for managing users, products, and overall system activities.	Technical knowledge, system management skills, administrative access

2.4. Operating Environment

The AgroSmart Marketplace system operates as a web-based application that requires an internet connection and a compatible web browser for access. The system is developed using modern web technologies to provide a reliable environment for buyers, farmers, and administrators.

The operating environment of the system includes the following:

- **Backend Framework:** Laravel
- **Programming Language:** PHP
- **Frontend Technologies:** HTML, Tailwind CSS, JavaScript, Alpine.js
- **Database Management System:** MySQL
- **Web Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge, and other modern browsers
- **Supported Devices:** Desktop computers, laptops, and internet-enabled devices
- **Development Tools:** Visual Studio Code, Composer, NPM, Laravel Artisan Server, MySQL Workbench
- **Network Requirement:** Stable internet connection

2.5. Assumptions and Dependencies

- Users have access to a stable internet connection for system access and communication.
- Buyers and farmers have basic computer or web browsing knowledge to use the system effectively.
- Users provide accurate product, pricing, and account-related information.
- The system depends on a compatible web browser for proper functionality.
- The system depends on Laravel, PHP, MySQL, and related development technologies for operation.
- The database server must function properly to store and retrieve system data.
- User authentication and role-based access control must operate correctly for secure access.
- Location-based features depend on valid district and city information provided by users.
- Demand analysis depends on user-generated activities such as searches, views, offers, and orders for accurate results.

3. Functional Requirements

3.1. User Registration and Authentication

Functional Requirements:

- The system shall allow buyers and farmers to register by entering required details such as name, email, password, role, district, city, and address.
- The system shall validate registration details before creating a user account.
- The system shall prevent duplicate registration using the same email address.
- The system shall allow registered users to log in using their email address and password.
- The system shall authenticate users before allowing access to protected pages.
- The system shall redirect authenticated users to the relevant dashboard based on their user role.
- The system shall allow users to log out securely.
- The system shall restrict public registration for administrator accounts.

3.2. Role-Based Dashboard Access

Functional Requirements:

- The system shall identify the role of the authenticated user after login.
- The system shall redirect buyers to the Buyer Dashboard.
- The system shall redirect farmers to the Farmer Dashboard.
- The system shall redirect administrator to the Admin Dashboard.
- The system shall restrict users from accessing dashboards that do not belong to their role.
- The system shall display role-specific menus, features, and dashboard content according to the logged-in user’s role.

3.3. Farmer Product Management

Functional Requirements:

- The system shall allow farmers to add new agricultural products.
- The system shall allow farmers to enter product details such as product name, category, quantity, price, minimum price, location, and product image.
- The system shall allow farmers to view their published products.
- The system shall allow farmers to update existing product details.
- The system shall allow farmers to remove or deactivate products that are no longer available.
- The system shall display farmer products to buyers through the product browsing section.
- The system shall allow farmers to manage product availability status.

3.4. Buyer Product Browsing and Search

Functional Requirements:

- The system shall allow buyers to view available agricultural products.
- The system shall display product details such as product name, price, farmer name, location, demand level, and availability.
- The system shall allow buyers to search for products using product name or related keywords.
- The system shall allow buyers to filter products by category, district or location, price range.
- The system shall allow buyers to filter products by demand level, such as high demand or low demand.
- The system shall allow buyers to view detailed information about a selected product.
- The system shall display recommended nearby products based on buyer location information.

3.5. Submit Offer Function

Functional Requirements:

- The system shall allow buyers to submit an offer for a selected product.
- The system shall display selected product details before the buyer submits an offer.
- The system shall allow buyers to enter an offer price.
- The system shall allow buyers to enter the required quantity.
- The system shall allow buyers to enter an optional note with the offer.
- The system shall validate offer details before submission.
- The system shall store submitted offer details for farmer review.
- The system shall notify or display the submitted offer to the relevant farmer.

3.6. Minimum Price Validation

Functional Requirements:

- Allow farmers to set a minimum price for each product.
- Compare the buyer's offer price with the product's minimum price.
- Restrict or reject offers below the farmer-defined minimum price.
- Display a warning message when a buyer enters an offer below the minimum price.
- Allow offer submission only when the offer price satisfies the minimum price rule.

3.7. Negotiation Management

Functional Requirements:

- Display submitted buyer offers to the relevant farmer.
- Allow farmers to view offer details such as buyer name, offer price, quantity, and note.
- Allow farmers to accept a suitable buyer offer.
- Allow farmers to reject unsuitable buyer offers.
- Allow farmers to respond with a counter-offer when needed.
- Update the offer status according to the farmer's decision.

- Notify or display the negotiation status to the buyer.

3.8. Order Management

Functional Requirements:

- Create an order after a farmer accepts a buyer offer.
- Display confirmed order details to the buyer and farmer.
- Store order details such as product, quantity, accepted price, buyer, farmer, and order status.
- Allow buyers to view their confirmed and completed orders.
- Allow farmers to view orders received from buyers
- Update order status according to the order progress.
- Mark orders as completed after the product exchange is finalized.

3.9. Location-Based Matching

Functional Requirements:

- Store district and city details during user registration.
- Display products based on farmer location information.
- Allow buyers to filter products by district or city.
- Show recommended nearby products according to the buyer's location.
- Display product location details to help buyers identify nearby farmers.
- Support location-based product browsing to improve buyer convenience.

3.10. Demand Analysis

Functional Requirements:

- Collect buyer activity data such as searches, views, offers, and orders.
- Analyze collected activity data to identify product demand levels.
- Categorize products as high-demand or low-demand based on demand analysis results.
- Display demand insights through a dashboard.

- Provide demand-related information to support farmer pricing and selling decisions.
- Update demand insights when new buyer activity data is available.

3.11. Smart Pricing Support

Functional Requirements:

- Allow farmers to enter product price and minimum price values.
- Use demand level information to support pricing decisions.
- Provide suggested price ranges based on system-defined pricing rules.
- Display pricing guidance to farmers when adding or updating products.
- Help farmers avoid setting prices below acceptable minimum values.
- Support fair and competitive pricing for agricultural products.

3.12. Review and Feedback Management

Functional Requirements:

- Allow buyers to submit reviews after completed orders.
- Allow buyers to provide ratings for farmers or products.
- Allow buyers to add comments about product quality and service experience.
- Display reviews and ratings for relevant products or farmers.
- Store review details such as buyer, farmer, product, rating, comment, and date.
- Allow administrators to monitor reviews and feedback for inappropriate content.

3.13. Admin Management

Functional Requirements:

- Allow administrators to log in using authorized admin accounts.
- Allow administrators to view registered buyers and farmers.
- Allow administrators to manage user accounts when required.
- Allow administrators to view and monitor product listings.
- Allow administrators to manage inappropriate or invalid product information.
- Allow administrators to manage system activities to maintain platform reliability.

4. Interface Requirements

4.1.User Interface Requirements

The user interface of AgroSmart Marketplace should be simple, responsive, and easy to use. It should allow buyers, farmers, and administrators to access their relevant features through clear dashboards, navigation menus, forms, product cards, and interactive components.

The user interface requirements are as follows:

- Provide login and registration interfaces for buyers and farmers.
- Provide role-based dashboards for buyers, farmers, and administrators.
- Include a sidebar and top navigation bar for easy access to main features.
- Display product information using clear product cards with image, name, price, location, farmer details, and demand level.
- Provide a browse products page with search and filtering options.
- Provide a submit offer popup that allows buyers to enter offer price, quantity, and an optional note.
- Provide an add review popup that allows buyers to submit a rating and feedback comment after a completed order.
- Provide product management forms for farmers to add, update, and manage agricultural products.
- Provide offer management interfaces for farmers to view, accept, reject, or respond to buyer offers.
- Display demand insights using charts, labels, or visual indicators.
- Provide admin interfaces to manage users, products, reviews, and platform activities.
- Provide a language selection option that allows users to choose Sinhala, Tamil, or English as their preferred interface language.
- Show validation, success, and error messages where required.
- Use clear labels, buttons, icons, and colors to guide users.
- Support responsive layouts for desktop, laptop, and modern web browsers.

4.2. Hardware Interface Requirements

The AgroSmart Marketplace system is a web-based application and does not require any special hardware devices to operate.

The hardware interface requirements are as follows:

- Support access through desktop computers, laptops, tablets, and smartphones.
- Require a keyboard and mouse or touch screen for user interaction.
- Require a display screen to view dashboards, forms, product details, and system messages.
- Require an internet-enabled device to access the web-based system.
- Require server hardware or hosting infrastructure to run the web application and database.
- Support image upload from user devices for product images where required.
- No special external hardware devices are required for normal system operation.

4.3. Software Interface Requirements

Software Component	Purpose
Laravel	Handles routing, backend logic, authentication flow, and system operations.
PHP	Supports server-side processing and backend functionality.
MySQL	Stores users, products, offers, orders, reviews, and activity-related data.
Blade Templates	Provides dynamic frontend page rendering
HTML and Tailwind CSS	Structures and styles the user interfaces.
JavaScript and Alpine.js	Supports interactive UI functions such as dropdowns, popups, and sidebar actions.
Laravel Jetstream and Fortify	Provides registration, login, logout, and authentication support.
Image Upload Handling	Allows product image uploads where required.

4.4. Communication Interface Requirements

The AgroSmart Marketplace system requires communication between the user's web browser, the application server, and the database server. Since the system is web-based, user requests and system responses are handled through standard web communication methods.

The communication interface requirements are as follows:

- Support communication between the web browser and the application server using HTTP/HTTPS protocols.
- Send user requests such as login, registration, product browsing, offer submission, and review submission to the server.
- Retrieve and display data such as product details, user information, offers, orders, and demand insights from the server.
- Support secure data transmission for authentication and user-related information.
- Communicate with the MySQL database to store, update, retrieve, and delete system data.
- Support image upload communication for agricultural product images where required.
- Provide appropriate system responses, including success messages, error messages, and validation messages.

5.1. Technology Constraints

Constraint Area	Description
Backend	The system is limited to Laravel and PHP for backend development.
Database	MySQL is used as the main database management system.
Frontend	Blade, HTML, Tailwind CSS, JavaScript, and Alpine.js are used for interface development.
Authentication	Laravel Jetstream and Fortify are used for authentication handling.
Platform	The system is limited to a web-based platform and does not include a mobile application.
Build Tools	Vite, Composer, and NPM are used for asset and dependency management.

5.2. Database Constraints

The database constraints are as follows:

- MySQL must be used as the main database management system.
- User details such as name, email, password, role, district, city, and address must be stored securely.
- User roles must be properly stored and managed to support Buyer, Farmer, and Admin access.
- Product records must include details such as product name, category, price, minimum price, quantity, location, availability, and product image.
- Buyer offer details such as offer price, quantity, optional note, buyer ID, product ID, and offer status must be stored.
- Order records must store accepted offer details, product information, buyer, farmer, quantity, accepted price, and order status.
- Review records must store buyer feedback, rating, comment, related product, farmer, and completed order details.
- Buyer activities such as searches, views, offers, and orders should be stored or tracked to support demand analysis.

- Relationships between users, products, offers, orders, and reviews must be maintained using proper primary keys and foreign keys.
- Duplicate email addresses must not be allowed during user registration.
- Sensitive data such as user passwords must be stored securely using hashing.
- The database should support backup and maintenance procedures to reduce the risk of data loss.

5.3. Security Constraints

The security constraints are as follows:

- User passwords must be stored securely using hashing.
- Only registered users should be allowed to access protected system pages.
- Role-based access control must be applied to separate Buyer, Farmer, and Admin functions.
- Buyers should not access farmer management functions, and farmers should not access buyer-only functions.
- Administrator functions must be restricted to authorized admin users only.
- Public registration for administrator accounts must not be allowed.
- User input must be validated to prevent invalid data and common security issues.
- Sensitive user information such as email, password, and location details must be protected.
- The system should prevent unauthorized access to product, offer, order, and review data.
- Secure logout must be provided to end user sessions properly.
- Database access should be restricted to authorized system operations only.
- Regular backup and maintenance procedures should be considered to reduce data loss and security risks.

6. Non-Functional Requirements

Non-functional requirements define the quality, performance, and operational standards of the AgroSmart Marketplace system. These requirements ensure that the system is efficient, secure, reliable, user-friendly, maintainable, and scalable.

6.1. Performance

The system should perform efficiently during normal user activities.

The Performance requirements are as follows:

- Support multiple buyers, farmers, and administrators at the same time without major delay.
- Load main pages such as login, dashboard, and browse products within a reasonable time.
- Process product browsing, searching, filtering, offer submission, and database queries efficiently.
- Optimize product images and dashboard data to improve page speed.

6.2. Security

The system should protect user data and prevent unauthorized access.

The security requirements are as follows:

- Store user passwords securely using hashing.
- Apply role-based access control for Buyer, Farmer, and Admin users.
- Prevent unauthorized access to protected pages and restricted functions.
- Validate user inputs to reduce invalid data and security risks.
- Restrict public registration for administrator accounts.

6.3. Usability

Usability defines how easily users can understand, access, and use the AgroSmart Marketplace system.

The usability requirements are as follows:

- Provide a clear and easy-to-use interface for all user roles.
- Use consistent navigation menus, buttons, labels, icons, and page layouts.
- Display understandable success, error, and validation messages.
- Allow users to access important functions with minimum steps.
- Support Sinhala, Tamil, and English language options to improve accessibility for local users.
- Provide responsive layouts for desktop, laptop, and modern web browsers.
- Require minimal training for users with basic web browsing knowledge.

6.4. Reliability

Reliability defines the ability of the AgroSmart Marketplace system to operate consistently and maintain accurate data during normal usage.

The reliability requirements are as follows:

- Handle common errors without crashing the system.
- Maintain accurate records for users, products, offers, orders, reviews, and demand-related activities.
- Ensure database relationships and data consistency between products, offers, orders, and users.
- Display appropriate error or validation messages when an operation cannot be completed.
- Support backup and recovery procedures to reduce the risk of data loss.
- Maintain stable operation during normal user activities such as login, product browsing, product management, and offer submission.

6.5. Maintainability

Maintainability defines how easily the AgroSmart Marketplace system can be updated, corrected, and improved in the future.

The maintainability requirements are as follows:

- Use clean and well-structured code to make future updates easier.
- Follow a modular design using reusable components such as layouts, forms, cards, and dashboards.
- Allow developers to fix errors and add new features without affecting existing system functions.
- Maintain proper documentation for system features, database structure, and development process.
- Organize frontend, backend, and database files clearly for easier understanding and maintenance.
- Support future improvements such as online payments, delivery management, and advanced demand analysis.

6.6. Scalability

The scalability requirements are as follows:

- Support an increased number of buyers, farmers, administrators, products, offers, and orders.
- Allow the database to store and manage additional product, user, offer, order, review, and activity data.
- Support future integration of online payment and delivery management features.
- Allow future development of a dedicated mobile application.
- Support future enhancement of smart pricing and demand analysis using advanced data analysis methods.
- Allow new modules and features to be added without affecting existing system functions.

7. Database Design Section

7.1. Database Objectives

The database is designed to:

- Store buyer, farmer, and administrator information securely.
- Manage agricultural product details published by farmers.
- Store buyer offer details for negotiation-based trading.
- Maintain order records after accepted offers.
- Store reviews and feedback after completed orders.
- Track buyer activities such as searches, views, offers, and orders for demand analysis.
- Reduce data duplication and maintain data consistency.
- Support fast data retrieval and updates.

7.2. Main Database Tables

1. Users Table

Stores all registered user information including buyers, farmers, and administrators.

Field Name	Data Type	Description
user_id	INT (PK)	Unique user identifier
full_name	VARCHAR (100)	User full name
phone_number	VARCHAR(15)	User contact number
password	VARCHAR(255)	Hashed user password
role	ENUM	Buyer, Farmer, or Admin
city	VARCHAR(100)	User city
created_at	TIMESTAMP	Account creation date

2. Products Table

Stores agricultural product details published by farmers.

Field Name	Data Type	Description
product_id	INT (PK)	Unique product identifier
farmer_id	INT (FK)	Farmer who published the product
product_name	VARCHAR(100)	Name of the agricultural product
category	VARCHAR(100)	Product category such as vegetables, fruits,
description	TEXT	Product description
price	DECIMAL(10,2)	Selling price per unit or kilogram
minimum_price	DECIMAL(10,2)	Minimum acceptable price set by the farmer
quantity	DECIMAL(10,2)	Available product quantity
unit	VARCHAR(20)	Product unit such as kg, bundle, or packet
district	VARCHAR(100)	Product location district
city	VARCHAR(100)	Product location city
product_image	VARCHAR(255)	Uploaded product image path
demand_level	ENUM	High Demand or Low Demand
availability_status	ENUM	Available or Unavailable

3. Offers Table

Stores buyer price offers submitted for agricultural products.

Field Name	Data Type	Description
offer_id	INT (PK)	Unique offer identifier
product_id	INT (FK)	Related product identifier
buyer_id	INT (FK)	Buyer who submitted the offer
offer_price	DECIMAL(10,2)	Price offered by the buyer
quantity	DECIMAL(10,2)	Quantity requested by the buyer
note	TEXT	Optional message from the buyer
status	ENUM	Offer status such as Pending, Accepted, Rejected, or Countered

4. Orders Table

Stores confirmed order details after a farmer accepts a buyer offer.

Field Name	Data Type	Description
order_id	INT (PK)	Unique order identifier
offer_id	INT (FK)	Related accepted offer
product_id	INT (FK)	Related product identifier
buyer_id	INT (FK)	Buyer who placed the order
farmer_id	INT (FK)	Farmer who owns the product
quantity	DECIMAL(10,2)	Ordered quantity
accepted_price	DECIMAL(10,2)	Final accepted price
total_amount	DECIMAL(10,2)	Total order amount
status	ENUM	Order status such as Pending, Confirmed, Ready, or Completed

5. Reviews Table

Stores buyer ratings and feedback after completed orders.

Field Name	Data Type	Description
review_id	INT (PK)	Unique review identifier
order_id	INT (FK)	Related completed order
buyer_id	INT (FK)	Buyer who submitted the review
farmer_id	INT (FK)	Farmer who received the review
product_id	INT (FK)	Product related to the review
rating	INT	Rating value given by the buyer
comment	TEXT	Optional feedback comment
created_at	TIMESTAMP	Review creation date

6. Demand Activities Table

Stores buyer activities used for demand analysis.

Field Name	Data Type	Description
activity_id	INT (PK)	Unique activity identifier
buyer_id	INT (FK)	Buyer who performed the activity
product_id	INT (FK)	Product related to the activity
activity_type	ENUM	Activity type such as Search, View, Offer, or Order
activity_date	TIMESTAMP	Date and time of the activity

7. System Activities Table

The system activities table records important administrative actions performed within the platform.

Field Name	Data Type	Description
activity_id	BIGINT UNSIGNED (PK)	Unique identifier of the administrative activity
user_id	BIGINT UNSIGNED (FK)	References the administrator who performed the action
action	VARCHAR(255)	Name of the administrative action
module	VARCHAR(255)	System module where the action occurred
description	TEXT	Detailed description of the activity
ip_address	VARCHAR(45), NULL	IP address used to perform the action
user_agent	TEXT, NULL	Browser and device information of the administrator

7.3. Primary Keys and Foreign Keys

Table Name	Primary Key	Foreign Key
Users	user_id	—
Products	product_id	farmer_id
Offers	offer_id	buyer_id, product_id
Orders	order_id	offer_id, buyer_id, farmer_id, product_id
Reviews	review_id	order_id, buyer_id, farmer_id, product_id
Demand Activities	activity_id	user_id, product_id
System Activities Table	activity_id	

7.4. Database Relationships

Relationship	Description
Farmer → Products	One farmer can publish many products.
Buyer → Offers	One buyer can submit many offers.
Product → Offers	One product can receive many buyer offers.
Offer → Order	An accepted offer can create one order.
Order → Review	A completed order can receive a review.
User → Demand Activities	One user can generate many demand-related activities.
Product → Demand Activities	One product can have many searches, views, offers, or order activities.

7.5. Database Constraints

The database constraints are as follows:

- Primary keys must uniquely identify each record.
- Foreign keys must maintain relationships between related tables.
- Required fields must not be empty.
- Duplicate email addresses must not be allowed.
- Passwords must be stored securely using hashing.
- Product records must be linked to the relevant farmer.

- Offer records must be linked to the relevant buyer and product.
- The farmer related to an offer is identified through the product record linked by product ID.
- Order records must be created only after an offer is accepted.
- Review records must be linked to completed orders.
- Demand activity data should be stored or tracked to support demand analysis.

7.6. Database Security

The database security requirements are as follows:

- Store user passwords using secure hashing.
- Apply role-based access control for Buyer, Farmer, and Admin users.
- Restrict unauthorized access to user, product, offer, order, and review data.
- Validate user inputs before storing data in the database.
- Maintain database backup and recovery procedures to reduce data loss.

7.7. Future Database Expansion

The database design should support future enhancements such as:

- Online payment records.
- Delivery and logistics management.
- Advanced smart pricing data.
- More detailed demand analysis records.
- Mobile application support.
- Should Support future multilingual content for Sinhala, Tamil, and English.

8. Testing Requirements Section

This section defines the testing requirements required to verify the correctness, reliability, usability, performance, and security of the AgroSmart Marketplace system. Testing will be carried out to ensure that the system functions meet the specified requirements and support the expected workflows of buyers, farmers, and administrators.

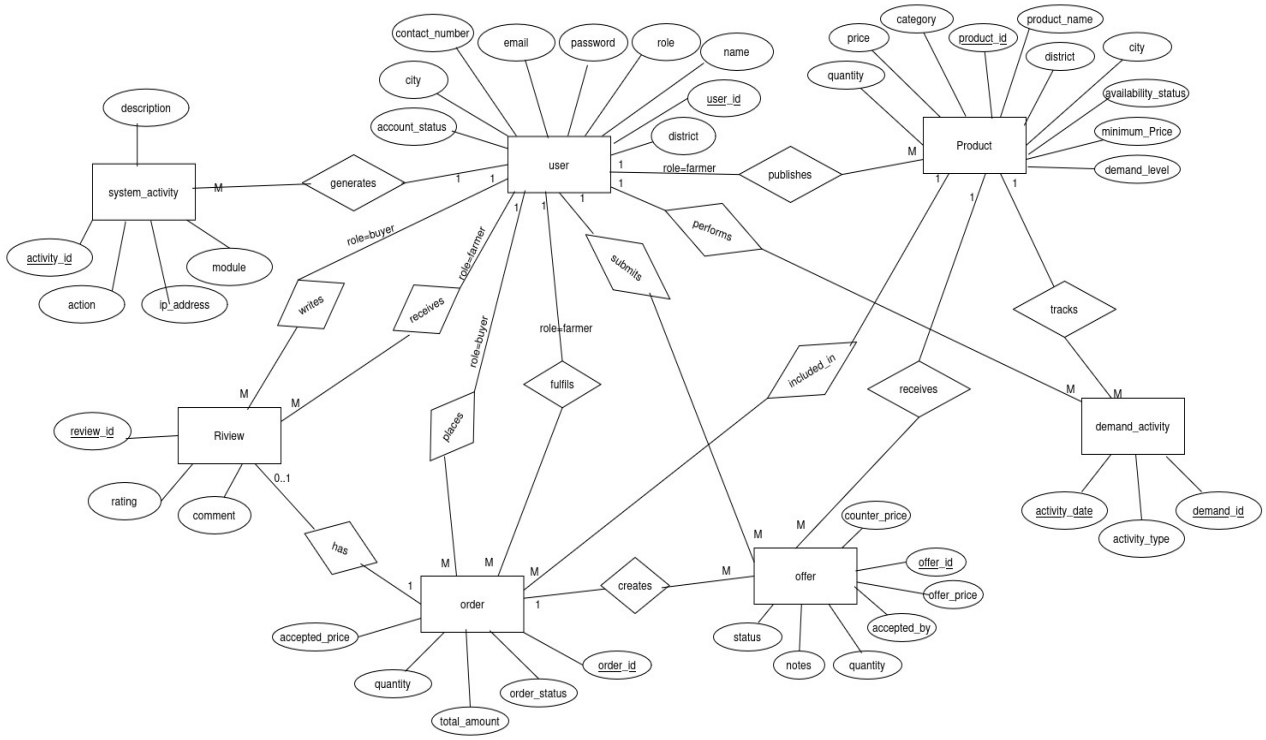
8.1. Testing Requirements Overview

Test Area	Purpose	Key Functions to Test	Expected Outcome
Unit Testing	Verify individual modules separately.	Registration, login, product add/update, offer submission, review submission.	Each module performs its intended function correctly.
Integration Testing	Verify communication between related modules.	Login with dashboard, product with database, offer with order, order with review.	Connected modules work together without errors.
System Testing	Verify the complete system as one application.	Buyer workflow, farmer workflow, admin workflow, demand insights, role access.	Complete system functions according to requirements.
User Acceptance Testing	Validate the system with expected user needs.	Buyer browsing and offers, farmer product and offer management, admin monitoring.	System is suitable for real-world use by target users.
Error Handling Testing	Check system response to invalid inputs.	Empty fields, duplicate email, invalid login, offer below minimum price.	Clear validation or error messages are displayed.
Performance Testing	Evaluate system speed and stability.	Page loading, product browsing, search/filter, offer submission, dashboard data.	System responds within a reasonable time under normal usage.

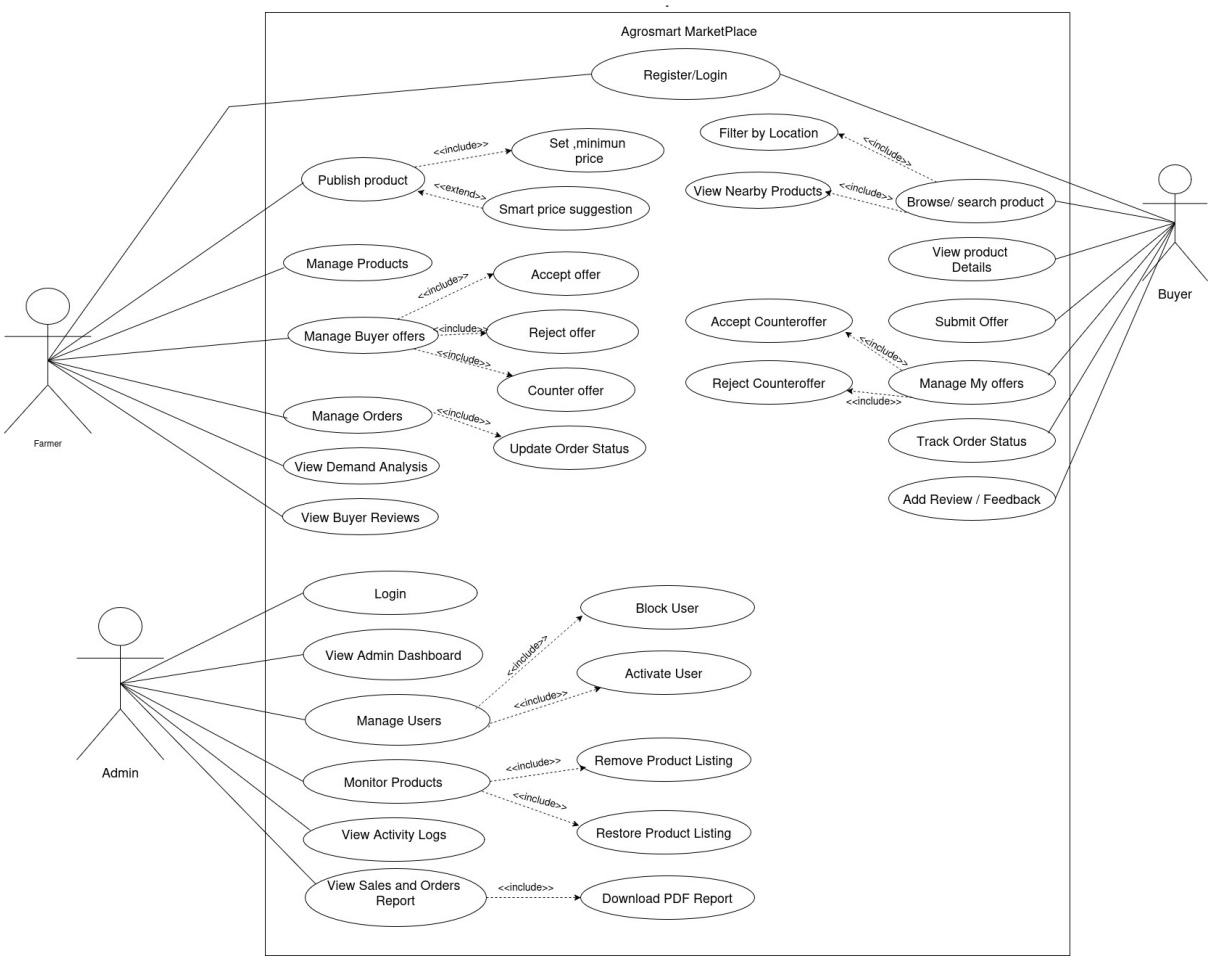
8.2. Sample Test Cases

Test ID	Test Area	Test Scenario	Expected Result
TC-01	Authentication	User logs in with valid credentials.	Correct role-based dashboard is displayed.
TC-02	Registration	User submits registration form with missing required fields.	Validation message is displayed.
TC-03	Product Management	Farmer adds valid product details.	Product is saved and displayed successfully.
TC-04	Product Browsing	Buyer searches or filters products.	Relevant products are displayed.
TC-05	Offer Submission	Buyer submits a valid offer for a product.	Offer is stored and shown to the farmer.
TC-06	Minimum Price Validation	Buyer submits an offer below minimum price.	Warning is displayed or offer is restricted.
TC-07	Order Management	Farmer accepts a buyer offer.	Order is created successfully.
TC-08	Review Management	Buyer adds a review after a completed order.	Review is saved successfully.
TC-09	Role Access	Buyer attempts to access farmer-only pages.	Access is denied.
TC-10	Demand Analysis	Buyer activities are processed for demand level.	High-demand and low-demand results are displayed.

9. ER Diagram


contact_number
email
password
role
name
city
user_id
account_status
district
user
description
system_activity
generates
activity_id
module
action
ip_address
role=buyer
role=farmer
writes
receives
submits
performs
publishes
role=farmer
Product
category
product_name
price
product_id
district
city
quantity
availability_status
minimum_Price
demand_level
tracks
role=buyer
role=farmer
fulfils
places
included_in
receives
review_id
Riview
rating
comment
0..1
has
demand_activity
activity_date
demand_id
activity_type
counter_price
offer
offer_id
offer_price
accepted_by
status
notes
quantity
creates
order
accepted_price
quantity
order_status
total_amount
order_id

10. Use case Diagrams



11. Appendices

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